

Brute Force Log Detection Lab

DVWA + Burp Suite + Splunk

Overview

This project demonstrates a basic SOC (Security Operations Center) workflow, where a brute force attack is simulated on a vulnerable web application and detected through log analysis.

The primary focus of this lab is capturing attack traffic and identifying it in logs using a SIEM.

Objectives

- Simulate a brute force attack on a login page
- Capture HTTP requests using Burp Suite
- Generate server-side logs
- Forward logs to Splunk
- Identify attack patterns through log analysis

Tools Used

- Burp Suite – Intercepting proxy & attack execution
- Splunk – Log ingestion and analysis
- DVWA – Vulnerable web application
- Splunk Universal Forwarder – Log forwarding
- Docker – Application deployment

Lab Architecture

```
Burp Browser -> Burp Suite -> DVWA -> Server Logs -> Forwarder -> Splunk
```

Setup Summary

Note: Sensitive information (IPs, file paths, system details) is intentionally omitted.

- DVWA deployed in a controlled environment
- Server logs exposed for monitoring
- Splunk configured to receive logs
- Forwarder configured to send logs to SIEM
- Burp Suite used with built-in browser

Attack Simulation

Step 1 — Access Target

- Open DVWA login page
- Use Burp browser for traffic interception

Step 2 — Capture Request

- Enable intercept in Burp
- Capture login request:

```
POST /login.php
```

Step 3 — Perform Brute Force

- Send request to Intruder
- Select the password field
- Launch attack using a wordlist

Attack Behavior

- Multiple login attempts generated
- Same endpoint repeatedly targeted
- High-frequency request pattern observed

Log Generation

During the attack, the server logs recorded:

- Repeated POST /login.php requests
- High number of authentication attempts
- Consistent request patterns

Log Detection in Splunk

Basic Search

```
index=* "POST /login.php"
```

Identify Repeated Attempts

```
index=* "POST /login.php"  
| stats count by clientip
```

Observed Pattern

- High number of requests from a single source
- Rapid sequence of login attempts

This indicates a brute force attack pattern.

Key Findings

- Brute force attacks can be identified through log patterns
- Even without alerts, abnormal behavior is visible in SIEM
- Repeated authentication attempts are a strong indicator

Limitations

This lab currently includes:

- Attack simulation

- Log ingestion
- Log analysis

But does NOT include:

- Alert automation
- Dashboard visualization
- Blocking mechanisms

Suggested Improvements

- Implement alert rules in Splunk
- Add time-based detection logic
- Build dashboards for visualization
- Apply rate limiting on login endpoint

Conclusion

This project demonstrates how a brute force attack can be simulated using Burp Suite, leaves identifiable traces in server logs, and can be detected using a SIEM like Splunk. It provides a foundation for SOC analysis and detection engineering.

Skills Gained

- Web request interception
- Brute force attack simulation
- Log analysis
- SIEM usage
- Understanding attack patterns

Disclaimer: This project was conducted in a controlled lab environment for educational purposes only.